



VOLKSWAGEN GROUP
DIGITAL SOLUTIONS [INDIA]

i.mobilathon 5.0

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H2S
HACK2SKILL

Team Details

- a. **Team name:** infinite recursion
- b. **Team leader name:** Agniv Dutta
- c. **Problem Statement:** Fatigue and stress significantly reduce driver alertness, increasing the risk of accidents. Using cabin video, steering behavior, and optional wearables, create a solution that detects drowsiness or stress levels and suggests safe, non-distracting interventions to keep drivers and passengers safe. (SafeDrive AI)

Brief about the idea

Real-Time Driver Wellness Monitoring

AI-powered vision technology that keeps drivers safe, alert, and in control



Drowsiness Detection

Identifies fatigue patterns before they become dangerous



Stress Recognition

Monitors psychological load in real-time



Distraction Alert

Tracks attention to the road instantly

How It Works

01

Computer Vision Analysis

Smartphone camera processes driver behavior continuously

02

On-Device AI Processing

All data stays private—zero cloud dependency, maximum security

03

Progressive Interventions

Smart alerts escalate from gentle nudges to urgent warnings

04

Driver Empowerment

Real-time feedback helps drivers make safer decisions

Who Benefits

- **Fleet Operators:** Reduce accidents, liability, insurance costs
- **Long-Distance Drivers:** Stay healthier, avoid fatigue-related incidents
- **Ride-Sharing Companies:** Enhance passenger safety, driver performance

Why It Matters

- Driver fatigue causes 20% of highway accidents
- Prevents costly incidents before they happen
- Privacy-first: no external data transmission
- Seamless integration with existing fleet systems

How different is it from any of the other existing ideas?

SafeDriver AI offers superior features over existing solutions.



High hardware cost



Cost-effective smartphone use



Limited detection accuracy



Multi-modal detection system



Privacy concerns



On-device data processing



Generic interventions



Adaptive interventions



Limited accessibility



Broad accessibility



Reactive detection



Predictive edge



Fixed algorithms



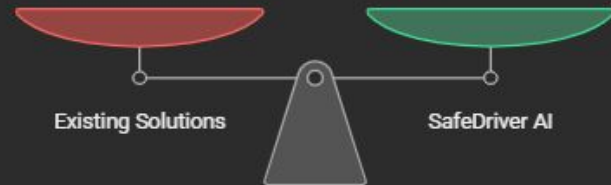
Adaptive learning



Isolated operation

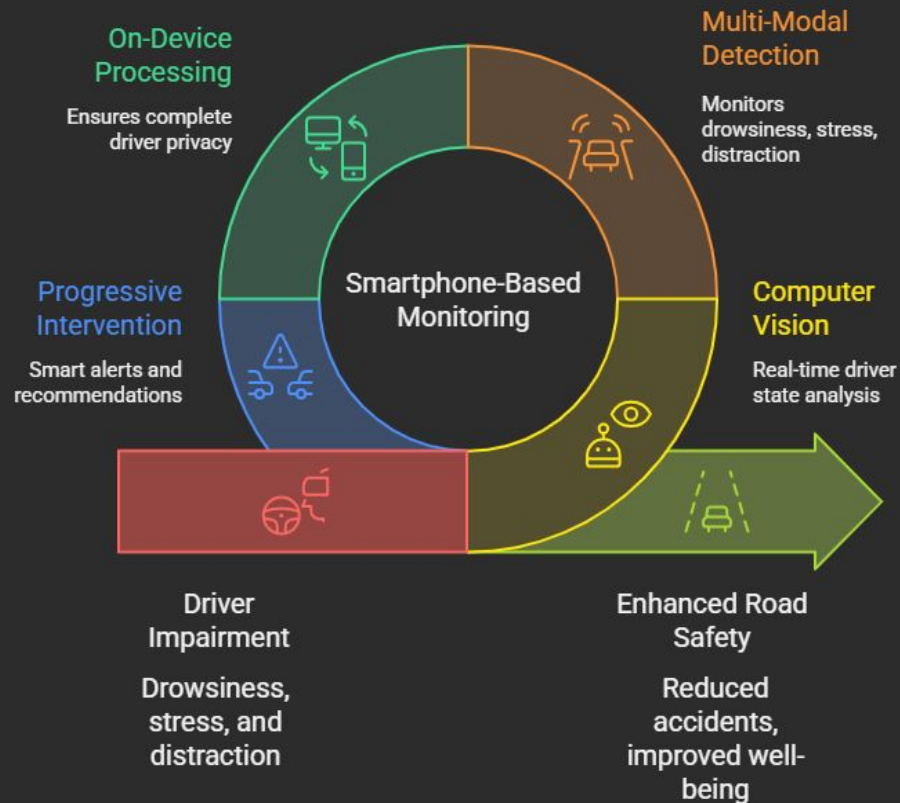


Fleet integrated



How will it be able to solve the problem?

Enhancing Road Safety with Driver Monitoring



USP of the proposed solution

Unique Selling Points

Zero Hardware Cost

Works with existing smartphones
—no device purchase required

Offline-First Operation

All processing on-device; works
even without internet

India-Focused

Adapted for Indian road conditions
and cultural context

Fleet-Ready

Scalable management portal for fleet operations

Open Architecture

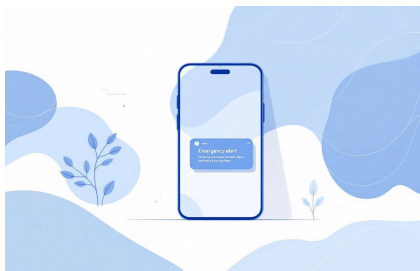
Integrates with Maps, smartwatches, car systems

List of features offered by the solution



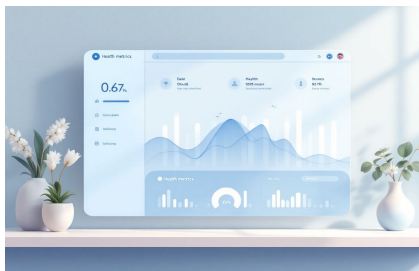
Fatigue Detection

Eye closure, yawns, head pose, blink rate analysis in real-time



Smart Alert System

Progressive alerts: vibration
→ audio → emergency
contact notification



Wellness Dashboard

Alertness score (0-100), trip
statistics, safety insights



Smart Interventions

Guided breathing exercises,
music playlists, rest stop
locator, privacy controls

Key Features: Intelligence & Integration

❑ Predictive Analytics

10-minute advance warning system using historical patterns and real-time data fusion

❑ Fleet Management Portal

Centralized dashboard for operators to monitor multiple drivers and trips

❑ Ecosystem Integrations

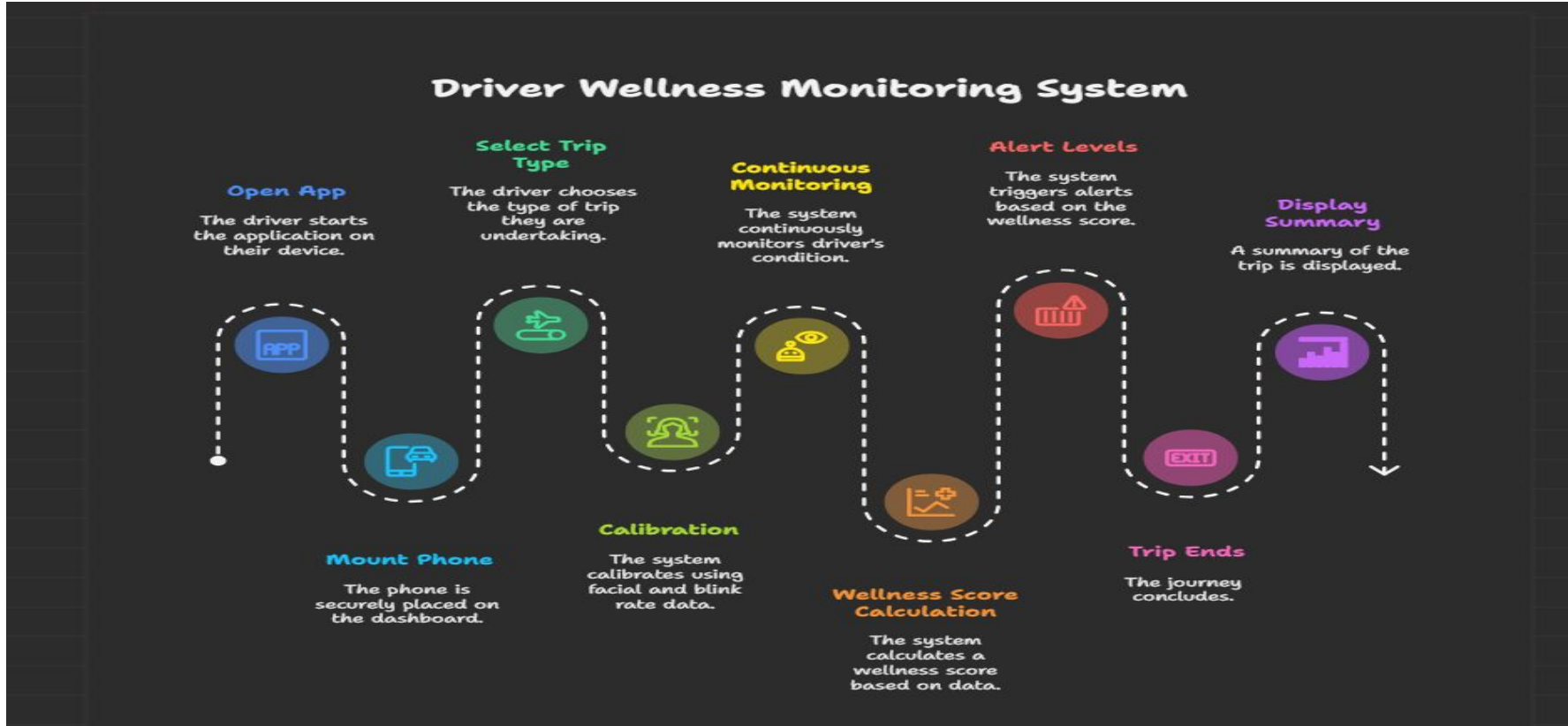
Google Maps, smartwatches, car Bluetooth, weather APIs, emergency services

❑ Social & Health Features

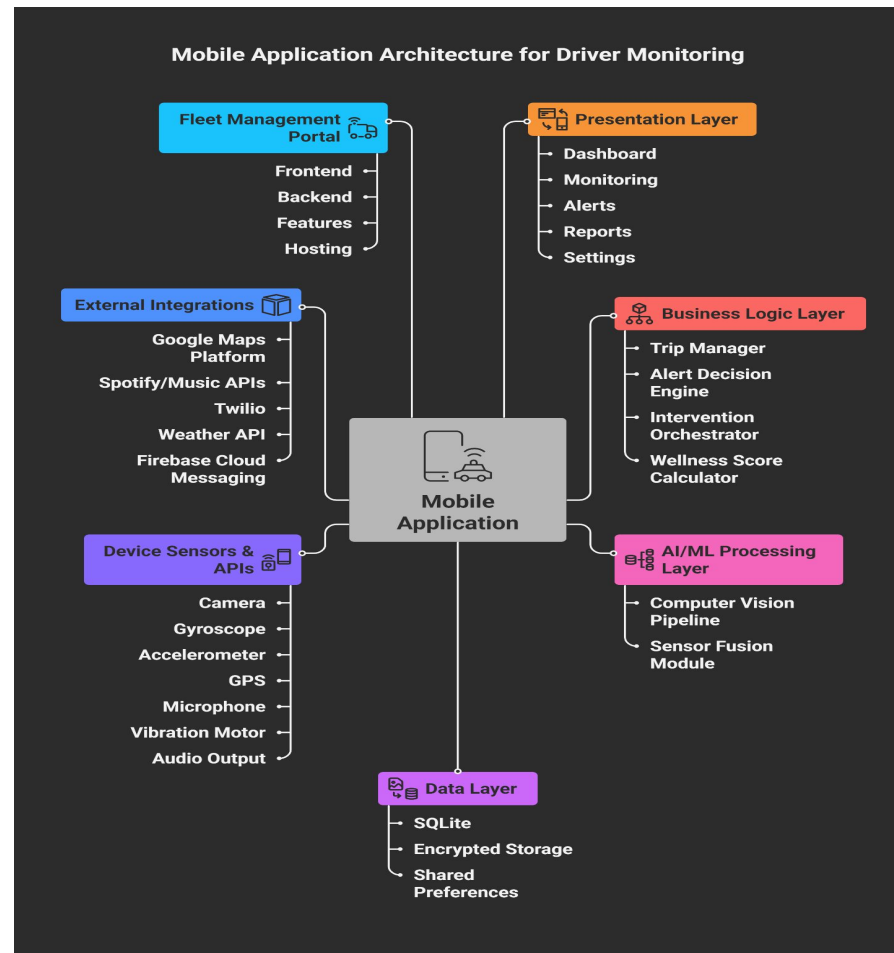
Safety leaderboards, sleep tracking, stress pattern analysis, performance incentives



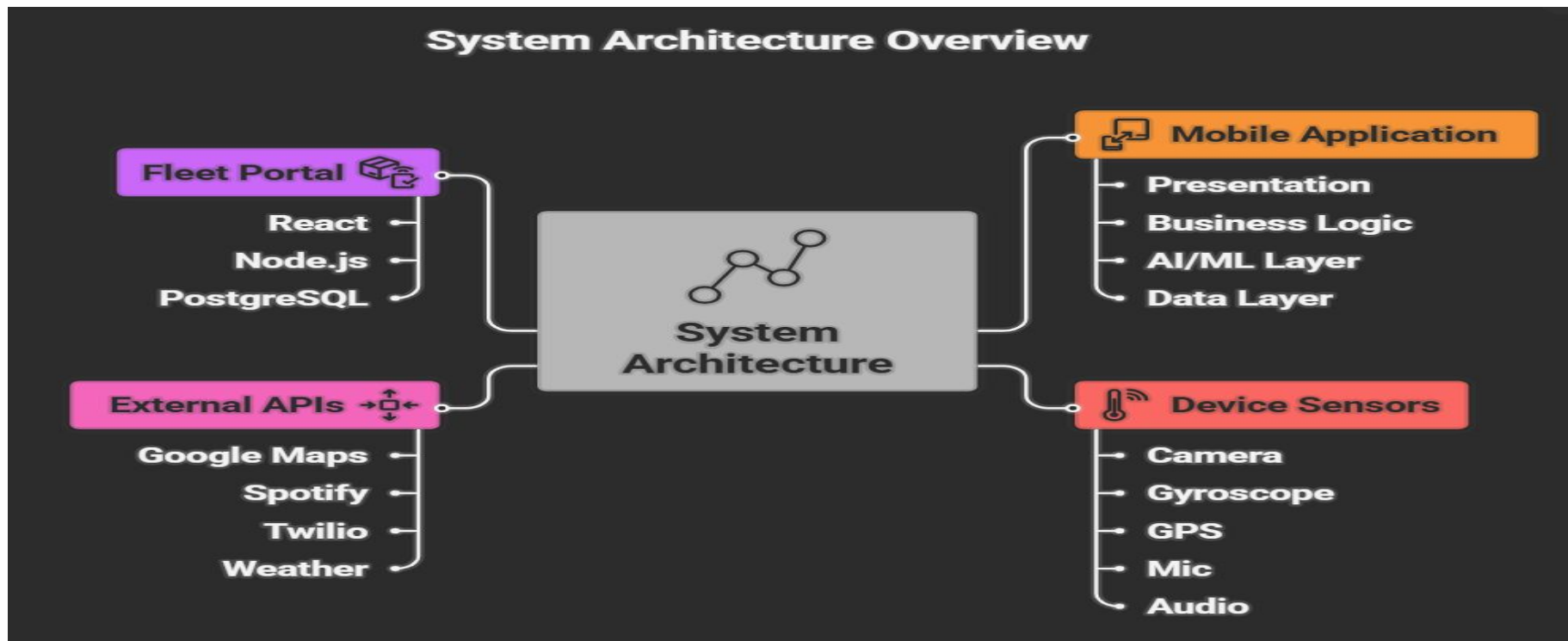
Process flow diagram or Use-case diagram



Wireframes/Mock diagrams of the proposed solution

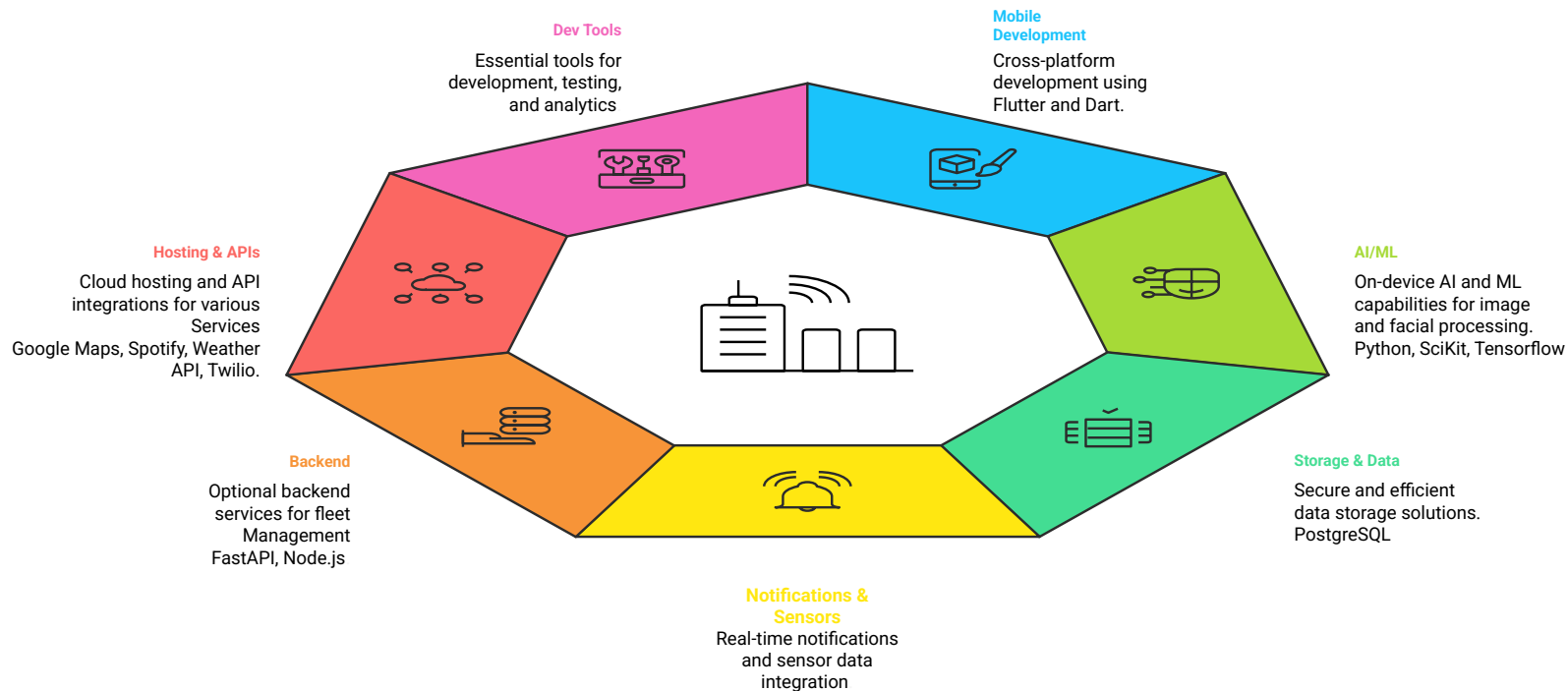


Architecture diagram of the proposed solution



Technologies to be used in the solution

Technology Stack Overview



Estimated implementation cost



Implementation & Impact

Development Cost

- Team & Engineers: ₹3,00,000
- ML Development: ₹1,50,000
- Design & Testing: ₹85,000
- Cloud & APIs: ₹25,000

Total: ₹5,70,000 (3 months)

Hackathon MVP: ₹0 (free tiers)

Expected Impact

- 25-30% accident reduction potential
- ₹2-3 lakhs/year savings per 10 drivers
- Break-even: 800 users or 50 fleet customers
- Target: 10,000 users Year 1
- 500+ lives saved annually

Revenue Model: Freemium (₹299/mo), Fleet (₹499/driver/mo), Insurance partnerships

Next Steps: Hackathon MVP → Pilot Program → Public Launch

References & Resources

SafeDrive AI — i.Mobilothon 5.0

Research & Data Sources

- **WHO Global Status Report (2023):** Fatigue-related accidents account for 20-30% of total crashes worldwide
- **NHTSA Research:** Comprehensive analysis of sleep deprivation's impact on driving performance
- **MoRTH India (2022):** 4.61 lakh accidents annually, with fatigue as a factor in 15-20% of highway incidents

Technology Stack

- **Google MediaPipe:** 468-point 3D facial landmark detection for real-time drowsiness analysis
- **TensorFlow Lite:** On-device ML for mobile-optimized inference and edge computing
- **Eye Aspect Ratio (EAR):** Soukupová & Čech algorithm for precise blink detection

Development & APIs

- **Frontend:** React 18, Tailwind CSS, Lucide React for responsive design
- **Integration:** Google Maps, Twilio emergency alerts, Spotify intervention system
- **Standards:** ISO 15007, SAE J3016, Euro NCAP compliance

SafeDrive AI: Project Resources

Access the SafeDrive AI platform and development repository to explore real-time monitoring capabilities for hospital operations and patient safety optimization.

Live Platform

Experience the SafeDrive AI MVP interface with interactive dashboards, real-time alerts, and operational insights designed for Indian healthcare environments.

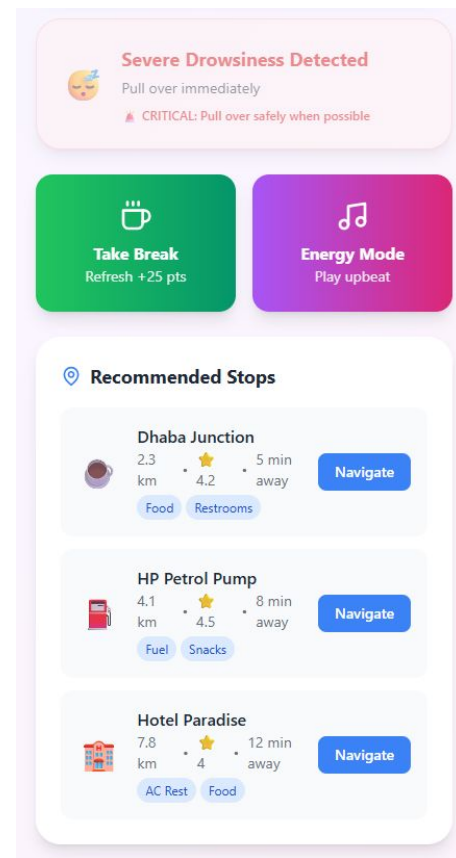
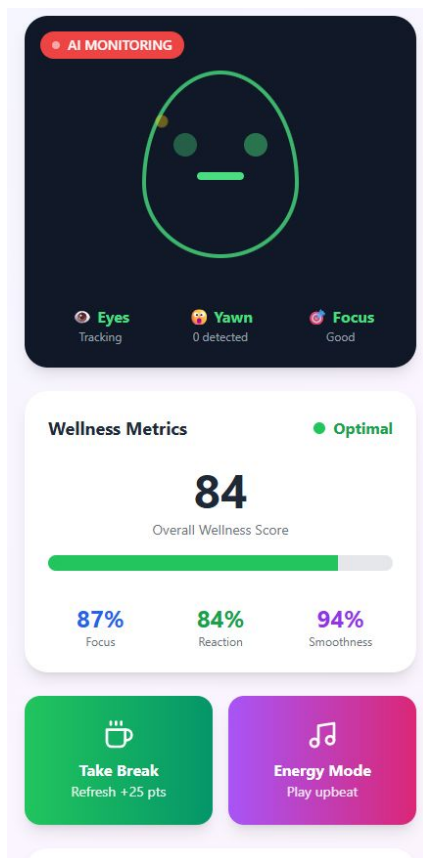
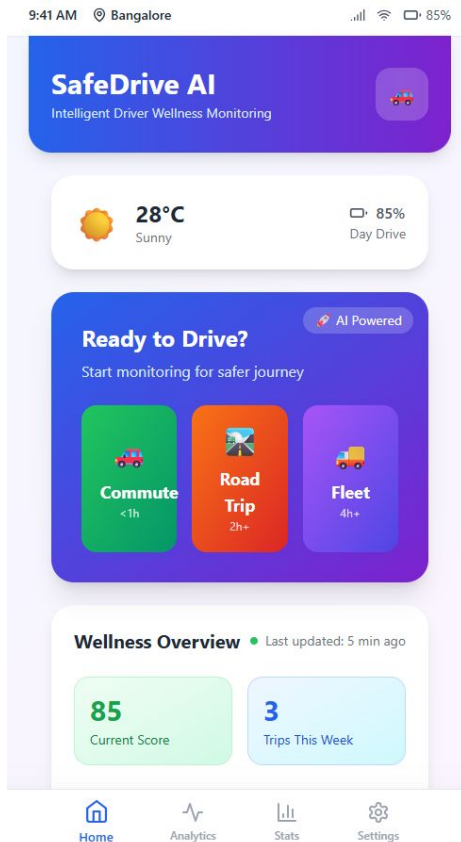
<https://safe-drive-ai-mvp-warj.vercel.app/>

Development Repository

Review the complete source code, technical architecture, and implementation details. Explore how SafeDrive AI integrates advanced monitoring with hospital workflows.

<https://github.com/AdityaC-07/SafeDrive-AI-MVP>

Screenshots of our MVP





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